Lesson 9.1: Corporate AI audits, ROI models and technology stack mapping

What you learn

- **Audit.** List the processes, who does them, hours spent, data involved and risks. Rank by value and ease.
- **ROI.** Annual saving = hours saved x hourly cost x weeks. Subtract build cost and yearly tools. Payback = build cost / monthly net saving.
- **Stack mapping.** Draw the tools, data and owners. Mark duplicates and gaps.
- **Honesty.** Label estimates as estimates and list assumptions.

Tools and resources (official sites, not checked live)

- NIST AI Risk Management Framework

ASSIGNMENT

Practice for lesson 9.1

Business problem: Calculate payback for one automation.

Steps

1. Run the tested ROI function.
2. Change the hours saved.
3. Explain the result to a manager.

Expected output: Payback in months and first-year ROI.

SOLUTION (ANSWER KEY)

PYTHON

```
def automation_roi(hours_saved_per_week, hourly_cost, weeks_per_year, build_cost,
monthly_tool_cost):
    annual_saving = hours_saved_per_week * hourly_cost * weeks_per_year
    annual_tool = monthly_tool_cost * 12
    first_year_net = annual_saving - build_cost - annual_tool
    net_per_year_after_build = annual_saving - annual_tool
    payback_months = build_cost / (net_per_year_after_build / 12) if net_per_year_after_build
> 0 else float("inf")
    roi_percent = first_year_net / (build_cost + annual_tool) * 100
    return {"annual_saving": annual_saving, "annual_tool": annual_tool,
            "first_year_net": first_year_net, "payback_months": round(payback_months, 1),
            "roi_percent": round(roi_percent, 1)}
```

With these example figures payback is 10 months and first-year ROI is 14.3 percent. All inputs are illustrative.

Check yourself

1. What is payback?
   Answer: Months to earn back the build cost.

2. Why list assumptions?
   Answer: So the client can challenge the numbers.

Lesson 9.1 of the **AI and Automation** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.